



Payment Gateway



- Payment gateway performs the important role in processing and authorizing the payment or transactions between customer and merchants.
- Facilitates online E- commerce transactions

Any transaction will be,

- **Authenticated** - Confirms the cardholder's identity
- **Authorization** - Ensures that a cardholder has the appropriate funds or credit needed to fulfill a transaction amount



Stakeholders in a Transaction

- **Merchant** - A business entity that accepts online payments in exchange for goods or services
- **Card Holder** - Obtains a card from an issuing bank, uses the account to pay for goods or services
- **Issuers** – Cardholder's bank, Approves or declines cardholder transactions, Releases funds to acquiring banks upon transaction approval
- **Acquirers** – Merchant's bank, Forwards transactions for authorization/authentication, settles amount to the merchant
- Most banks will have acquirer and Issuer Arm.
- **Card Networks** - Card networks pass information between the merchant's acquiring bank and an issuing bank or card issuer to decide whether you can make a purchase and to facilitate the purchase. Card Networks, (Visa, Mastercard, Rupay, Amex etc.) do not issue cards directly to the public but rather through the issuer bank.



How does a transaction work?

- **Step 1:** A customer will place an order by clicking on Pay in the merchant website.
- **Step 2:** Merchant securely transfers order information to the payment gateway. Payment Gateway initiates Authentication
- **Step 3:** After the authentication process is successful, the transaction is then authorized or declined by the issuing bank
- **Step 4:** Payment gateway sends a message to the merchant accordingly.
- **Step 5:** The bank settles the money with the payment gateway and then the payment gateway settles the money to the merchant minus the transaction fee.
- Acquirer PG makes money by levying a transaction fee.



Tech Stack and Good to knows

- Spring 3 or 4 or 5
 - JDK 1.7 or 1.8
 - Oracle SQL Database
 - JSP
 - Javascript
 - HTML
 - CSS
 - Bootstrap
 - JQuery
 - Servers used Tomcat, Weblogic, JBOSS, OHS, IIS etc,
-
- ISO 8583
 - Encryption and Hashing Algorithms, SHA-1, AES, DES etc.
 - SSL/TLS Handshakes , Certificates , Signature signing and verification
 - Tools like Wireshark for network debugging



Standalone VS MAPS Integrated PG

Standalone

- Has Merchant, Bank and Admin portal for Terminal and Merchant Configuration.
- Portals provide features for Risk Management, Transaction management, Fee Management and Reports
- Transaction Processing
- HDFC, CBO, NBO etc.

MAPS Integrated

- Only transaction processing
- Payment Gateway integrated with FSS MAPS application
- Settlement and Configuration taken care by MAPS, data from MAPS will be replicated to PG
- BRAC, BOB, SBI etc.



How is transaction sent to PG?

- Tranportal TCP/IP
- Tranportal HTTP
- Hosted TCP/IP
- Hosted HTTP
- Rest

- **Tranportal / Merchant Hosted**
 - Card Details collected by the merchant and sent in the request
 - Card Number
 - CVV
 - Exp Date

- **Hosted / Bank Hosted**
 - Card Details collected by Payment Gateway via Payment Page.
 - Card holder enters the card details directly.



Tranportal / Merchant Hosted Request



```
<request><type>C</type><langid>USA</langid><member>Test  
Test</member><amt>100.00</amt><currencycode>356</currencycode><action>1</action><trackid>16997162  
257</trackid><udf31>en-GB|24|648|360|-  
330|true|true</udf31>><card>4868946789345913</card><cvv2>289</cvv2><errorURL>https://api.razorpay.co  
m</errorURL>  
<responseURL>https://api.razorpay.com</responseURL>  
<expyear>2023</expyear><expmonth>06</expmonth><id>2859</id><password>***</password></request>
```

- type – Card/Payment Type
- langid – Language ID
- member – Card Holder Name
- amt – Transaction Amount
- currencycode – Currency to be used in the transaction
- trackid – Unique ID created by the merchant
- card – Card Number
- cvv2 – CVV of the Card
- expyear – Expiry Year of the Card
- expmonth – Expiry Month of the Card
- id – Unique ID assigned by PG to a merchant
- password – Password assigned by PG to a merchant
- ErrorURL – End point to which the Error response must be sent to
- Response URL – End point to which Transaction Response must be sent to



Tranportal / Merchant Hosted Response

- `<result>NOT CAPTURED</result><ref>307320011857</ref><postdate>2023-03-14
10:29:32.407</postdate><payid>101202307334975724</payid><tranid>202307365013987</tr
anid><amt>5790.51</amt><trackid>108571403231024660</trackid><udf31>en-
US|24|728|1366|-
330|true|true</udf31><type>Crebit_Card</type><card>552365*****7323</card>`
- result – Transaction Status
- ref – RRN number used for settlement with interchange and issuer, Unique ID generated by PG
- postdate – Transaction Date
- payid – Payment ID , Unique ID generated by PG for a transaction
- tranid – Tran ID, Unique ID generated by PG for a settlement



Hosted / Bank Hosted Transaction Request / Response



Request

- amt=900&action=1&responseURL=https://digitalsatbara.mahabhumi.gov.in/dslr/satbara/PaymentSuccessBOB&errorURL=https://digitalsatbara.mahabhumi.gov.in/dslr/satbara/PaymentSuccessBOB&trackid=23031700919¤cycode=356&langid=USA&udf6=Commissioner

Response

- paymentid=101202307687650273&result=IPAY0200007 - UPI Transaction Timed out.&custid=&postdate=17/03/2023 09:21:16&trackid=23031700919&udf6=Commissioner and Director Land Patwari&udf7=Customer name&udf8=customer@abc.com&udf9=9999999999&udf10=customer address&udf11=900&udf12=Any other Txn details as may be required by Merchant&udf13=mahabhumi.gov.in&amt=900.0



Rest API Request/ Response

Requests:

- [{"password":"TEST123456@","trackid":"169996200000337815","transId":"169996200000337815","udf5":"TrackID","action":"8","amt":"0.003","id":"ipay217435007025"}]

Response:

- [{"tranId":202301193523137,"date":"0111","authCode":"204596","trackId":"230100149210595948","udf5":"it@bima.om","amt":"13.655","udf3":"it@bima.om","udf4":"saifa al bhutta","udf1":"968/95542930","udf2":"TI","result":"CAPTURED","ref":"301180000000","paymentId":600202301106442310}]



Action Codes

- Decides which transaction is being initiated
- Type of Transactions

Action Code	Transaction
1	Purchase
2	Credit
3	Void Purchase
4	Auth
5	Capture
6	Void Credit
7	Void Capture
8	Inquiry
9	Void Auth



HTTP vs TCP IP

- HTTP – Browser based redirection, Encrypted
- TCP IP – Server to Server Calls, Usually in Plain text

Encryption Used

- Triple DES - Data Encryption Standard/AES - Advanced Encryption Standard
- AES/DES are Ciphers, an algorithm for performing encryption or decryption
- Both are symmetric ciphers – Uses on key for both encryption and decryption



Encrypted Request/Response - Sample

- http://10.44.6.135:8090/IPGBOB/PaymentHTTP.htm?param=paymentInit&trandata=86438D8F3ACC8210C6ECB266E16089DBC5141C6A765761BF84CD51A15FF230D2D0FBE2C1ADC33B6BEFB9D63A29B8181FA08F0AE66A65F5DE86AFA1438D11001586F911B4644400797AF18867235E8FF042990E14121848A90F27034F0FEC9CBEBBA2823057481EA3161F59FEF69023D959F567FC494CCA055BA1C5C08FAA261FA40F408907C9A51CD87368F6F413515EFA4CCCB2FA3DD4011463A037A6F4DDF2E9E04F3486A27B9C1AF069FF1446BCB22A9FD68AC44C0CA9BEC20D6088F1FD87E3B27ACD3C442F6B0D87B0C3D0A7B6FB216FA71A51BC9E7D496CF6D9610F7DB1E086F71BCFD97748D995167089536956ACF982416B4BEF3580B49936352369586F1578F93005E031FD92EC62F2B54D09DDAA90275A50AF789039FF7C63C4FF52AE6C0E3C078C2CEB02524E21CFAA84F67C88A3FBECDC82214ECE1B1BB9E45E816164210947F59720C8CCD8CD78B6A7501&errorURL=http://10.44.6.135:8090/DEV_FSSiPay2.1.0_Merchant_Demo/jsp/fss/HostedPaymentResultHTTP.jsp&responseURL=http://10.44.6.135:8090/DEV_FSSiPay2.1.0_Merchant_Demo/jsp/fss/HostedPaymentResultHTTP.jsp&tranportalId=2410



PG - Plugins

- Usually provided in JAVA , .NET and PHP
- Plugins aid in the integration of the merchants with Payment Gateway.
- To integrate, the following will be provided to the merchant by PG
 - The integration kit, JAR for Java, DLL for PHP/.NET
 - Resource File
 - Keystore File
 - Terminal Alias Name



Terminal and Merchant

- Merchants are onboarded with a Unique Merchant ID
- Each merchant may have one more Terminals.
- Each terminal is assigned a unique Terminal ID
- Tranportal ID , Password and Terminal Alias name created for each Terminal
- A merchant will have one resource file only with Tranportal ID and password of multiple terminals
- Terminal Alias name used to extract the Tranportal ID and Password of the terminal in Use.
- For encryption transactions, terminal resource key is generated and stored against the merchant



Resource/Keystore files

- resource.cgn and keystore.bin is provided to the merchant in zip folder.
- Keystore.bin - Contains the key to decrypt the resource.cgn
- Resource.cgn - Contains the secure data to be used in the transaction

```
<?xml version="1.0" encoding="UTF-8"?><terminal><id>425</id><password>hfjhfj465###</password><resourceKey>721615095763721615496776</resourceKey><webaddress>https://paymentgateway.com</webaddress></terminal>
```

- id – Tranportal ID, Unique ID assigned to the merchant
- password - Tranportal Password, password set for the merchant
- resourceKey - Key to be used for encryption/decryption of transaction requests/response
- webaddress - Web address of the Payment Gateway



Onus vs Offus Transactions

- Onus - Acquirer and Issuer is of the same bank
- Offus - Acquirer and Issuer is of the different bank

ONUS

- Example, a HDFC cardholder buys a product from a merchant integrated with the HDFC payment gateway.
- Authentication and Authorization could be done without interchange within the bank network

OFFUS

- Example, a SBI cardholder buys a product from a merchant integrated with the HDFC payment gateway.
- Interchange involvement is required



Card Based Transactions

- All Card based transactions will be
 - Authenticated – To confirm the cardholder
 - Authorized – To debit money from card holder's account
- For the successful card - based transactions below details are required
 - Card Number
 - Exp Date
 - CVV (Optional)
 - OTP (Optional)



OTP based Authentication

Stakeholders

- Card Network's DS (Directory Server)
- Issuer's ACS (Access Control Server)
- Acquirer PG's MPI (Merchant Server Plug-In)

Two protocols used,

- 3 D S – Three Domain Secure
- 3DS1 - Introduced in 2001
 - VEREQ/VERES
 - PAREQ/PARES
- 3DS2 – Introduced in 2015
 - PREQ/PRES - Preparation Request/Response
 - AREQ/ARES - Authentication Req/Res
 - CREQ/CRES - Challenge Req/Res
 - RREQ/RRES - Results Req/Res.
- Most implementations currently run in both 3DS1 and 3DS2



3D Secure 1.0

- Authentication protocol for card - based transactions
- OTP based authentication
- XML Based

API

VEREQ/ VERES – Verify Enrollment Request/Response

- Sent to Directory Server of the Card Network
- Used to check whether the card number is enrolled in 3DS1
- To get the endpoint of the issuer where PAREQ needs to be sent
- Server to Server call

PAREQ/PARES - Payer Authentication Request/Response

- Sent to Issuer
- OTP generation and validation
- Redirection based API



3D Secure 1.0 HTTP

1. Acquirer PG generates VEREQ and sends it via S2S call to the directory server of the card network
2. DS validates the request received from Acquirer, checks if merchant and acquirer is onboarded and if card number is enrolled with the card network
3. DS sends the VEREQ to Issuer ACS
4. ACS checks if the bin is actively enrolled in 3DS1 and sends the VERES to DS
5. DS sends VERES to PG
6. If VERES is successful, PG generates PAREQ and is redirected via browser used by the end user to the Issuer ACS
7. ACS generates and sends OTP to cardholder
8. Issuer renders the OTP page on the screen
9. Cardholder enters the OTP sent to his mobile phone
10. ACS validates the OTP
11. PARES is sent to the Acquiring PG by the Issuer ACS



3D Secure 1.0 TCP IP

1. Acquirer PG generates VEREQ and sends it via S2S call to the directory server of the card network
2. DS validates the request received from Acquirer, checks if merchant and acquirer is onboarded and if card number is enrolled with the card network
3. DS sends the VEREQ to Issuer ACS
4. ACS checks if the bin is actively enrolled in 3DS1 and sends the VERES to DS
5. DS sends VERES to PG
6. If VERES is successful, PG generates PAREQ and is sent to the merchant in a TCP IP API call
7. Merchant redirects PAREQ to Issuer ACS via browser used by the end user
8. ACS generates and sends OTP to cardholder
9. Issuer renders the OTP page on the screen
10. Cardholder enters the OTP sent to his mobile phone
11. ACS validates the OTP
12. PARES is redirected to the Merchant by the Issuer ACS
13. Merchant sends the PARES to PG via a TCP IP call



VEREQ - Verify Enrollment Request

- `<?xml version="1.0" encoding="UTF-8"?><ThreeDSecure><Message id="VEREQ202307687652568"><VEReq><version>1.0.2</version><pan>485498*****3752</pan><Merchant><acqBIN>403274</acqBIN><merID>2000000000000035</merID></Merchant><Browser><accept>text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8</accept><userAgent>Mozilla/5.0 (iPhone; CPU iPhone OS 16_2 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko)</userAgent></Browser></VEReq></Message></ThreeDSecure>`
- Message ID – Unique ID generated for every VEREQ request
- Version – Protocol Version(Usually constant)
- PAN – Card Number
- ACQ Bin – Bank Identification Number used to identify the acquirer constant
- MerID – Merchant ID
- Browser Accept and User agent – Details about the end user's browser, Collected during redirection for HTTP/



VERES - Verify Enrollment Response

Successful

```
<?xml version="1.0" encoding="UTF-8"?>
<ThreeDSecure><Message
id="VEREQ202307687652568"><VERes><version>1.0.2</version><CH><enrolled>Y</enrolled><
acctID>8sL8yU8IU4mG31679024714254</acctID></CH><url>https://netsafe.hdfcbank.com/ACSW
eb/com.enstage.entransact.servers.AccessControlServerSSL?ty=V</url><protocol>ThreeDSecure
</protocol></VERes></Message></ThreeDSecure>
```

Failure

```
<ThreeDSecure><Message
id="VEREQ202307687639727"><VERes><version>1.0.2</version><CH><enrolled>N</enrolled><
/CH><protocol>ThreeDSecure</protocol></VERes></Message></ThreeDSecure>
```

- version – 3DS version
- Enrolled – Y/N Indicator on whether card number is enrolled in 3DS1 or not
- AcctID - Cardholder Account Identifier
- url - Issuer endpoint to which PAREQ has to be sent



PAREQ – Payer Authentication Request

- `<?xml version="1.0" encoding="UTF-8"?><ThreeDSecure><Message
id="VEREQVEREQ202307687652568"><PAREq><version>1.0.2</version><Merchant><acqBIN>403274</acqBIN><merID>2000000000000035</merID><name>Paytm Payments
Utility P</name><country>356</country><url>https://ipg.bobgateway.com</url></Merchant><P
urchase><xid>PxW/Ft1hwh6Y2beTGYUjKDkdM0=</xid><date>20230317
09:14:57</date><amount>INR1,280.30</amount><purchAmount>128030</purchAmount><curr
ency>356</currency><exponent>2</exponent></Purchase><CH><acctID>8sL8yU8lU4mG3167
9024714254</acctID><expiry>2401</expiry></CH></PAREq></Message></ThreeDSecure>`
- Name – Merchant Name
- Country – Country Code
- URL – PG URL
- XID – Generated by PG, Transaction Identifier
- Amount/ Purch Amount – Transaction Amount
- Exponent – Exponent for Amount
- Acct ID – Received in VERES
- Expiry – Card Expiry date

PARES – Payer Authentication Response



- `<ThreeDSecure><Message id="VEREQVEREQ202307687652568"><PAREs id="PAREs4013850047"><version>1.0.2</version><Merchant><acqBIN>403274</acqBIN><merID>2000000000000035</merID></Merchant><Purchase><xid>PxW/Ft1hwh6Y2beTGYYUjKDkdM0=</xid><date>2023031709:14:57</date><purchAmount>128030</purchAmount><currency>356</currency><exponent>2</exponent></Purchase><pan>000000*****3752</pan><TX><time>2023031703:45:17</time><status>Y</status><caw>AAABBJGQGUGZZgTmUiZAZAAAAAAAAA=</caw><eci>05</eci><cawAlgorithm>2</cawAlgorithm></TX></PAREs>`
- PAN – Card Number
- Status – Decides whether authentication is successful
- ECI - Electronic Commerce Indicator (ECI)
- CAVV – Cardholder Authentication Value
- CAVV Algorithm

ECI Electronic Commerce Indicator (ECI)

Visa

- **ECI 05:** 3DS authentication was successful; transactions are secured by 3DS.
- **ECI 06:** authentication was attempted but was not or could not be completed; possible reasons being either the card or its Issuing Bank has yet to participate in 3DS.
- **ECI 07:** 3DS authentication is either failed or could not be attempted; possible reasons being both card and Issuing Bank are not secured by 3DS, technical errors, or improper configuration.

Mastercard

- **ECI 02:** 3DS authentication is successful; both card and Issuing Bank are secured by 3DS.
- **ECI 01:** 3DS authentication was attempted but was not or could not be completed; possible reasons being either the card or its Issuing Bank has yet to participate in 3DS, or cardholder ran out of time to authorize.
- **ECI 00:** 3DS authentication is either failed or could not be attempted; possible reasons being both card and Issuing Bank are not secured by 3DS, technical errors, or improper configuration.



CAVV

- Generated by the issuer
- Sent in authorization
- This can be proof of a fully authenticated or an attempted authentication transaction

CAVV Algorithm

The CAVV algorithm field in the PAREs message indicates the algorithm used to create the cryptogram contained in the CAVV field.

XID

The transaction identifier (XID) is a unique tracking number sent to the ACS in the Payer Authentication Request (PAREq) message.



PARES – Signature Verification

- All PAREs messages returned to the PG are digitally signed by the associated cardholder's issuer ACS using certificates provided by Interchange
- PG is required to validate the signature prior to extracting CAVV and sending it for authorization
- The CAVV value in the PAREs must be considered unusable if the signature validation process fails.
- Use the 3-D Secure Root Certificate to validate the cryptographic signature in the PAREs message to ensure its authenticity and integrity

To Verify the signature

- The public key of the party that signed the data.
- The digital signature.
- The data that was signed.



Authorization – ISO 8583

Responsible for debiting the amount from the cardholder's account

Stakeholders

- PG – Acquirer
- Base24 Switch – Acquirer
- Card Networks – Mastercard/Visa
- Issuer Switch

Onus vs Offus Authorization

Onus

PG --> Base24 Switch

Offus

PG --> Base24 Switch --> Card Network --> Issuer (HDFC, SBI etc)

or

PG --> Card Network --> Issuer (BOB)



- International Messaging Standard for Authorization – Retails Payments
- TCP /IP

Contains

- Header – Network Specific data
- Message Type Indicator Version – Version of ISO Bit Map 01000
- BitMap – What are the Data Elements set ?
 - Primary Bit Map (1-64)
 - Secondary Bit Map (64-128)
- Data Elements (1 – 128)

Message type

- 200/210 - Debit Purchase
- 220/230 - Refund
- 420/430 - Reversal
- 100/110 - Auth

Data element Key	Value
1	DE 1
2	4000 0000 0000 0002 (Card Number)
3	DE 3
4	DE 4
5	DE 5
.....	...



ISO Response - 100

- Message Type: 0100
- Bit Map: 723C440188E18008
- Bit #2 Primary Account Number:: 16 5535830507145293
- Bit #3 Processing Code:: 000000
- Bit #4 Transaction Amount:: 000001548900
- Bit #7 Transmission DateTime:: 0214095844
- Bit #11 Systems Trace Audit Number:: 457778
- Bit #12 Local Transaction Time:: 152846
- Bit #13 Local Transaction Date:: 0214
- Bit #14 Expiration Date:: 2412
- Bit #18 Merchant Type:: 5999
- Bit #22 Point of Service Data Code:: 810
- Bit #32 Acquiring Institution Identification Code:: 06018605
- Bit #33 Forwarding Institution Identification Code:: 06018605
- Bit #37 Retrieval Reference Number:: 304510028931
- Bit #41 Card Acceptor Terminal Identification:: 30000791
- Bit #42 Card Acceptor Identification Code:: 2000000000000582
- Bit #43 Card Acceptor Name/Location:: E-Commerce RP Bengaluru IND
- Bit #48 Additional Data, Private::
069T420701032124328jPmKUuKa+k/aCBEC65wEA5gAAAA=6105000016605010119203847
- Bit #49 Transaction Currency Code:: 356
- Bit #61 POS Data:: 0251025100006000356000000000



ISO Response - 110

- Message Type: 0110
- Bit Map: 7ee300018e81e102
- Bit #2 Primary Account Number 5535830507145293
- Bit #3 Processing Code 000000
- Bit #4 Transaction Amount 000001548900
- Bit #5 Settlement Amount 000000018744
- Bit #6 Cardholder Billing Amount 000001548900
- Bit #7 Transmission DateTime 0214095844
- Bit #9 Settlement Conversion Rate 70121015
- Bit #10 Cardholder Billing Conversion Rate 61000000
- Bit #11 Systems Trace Audit Number 457778
- Bit #15 Settlement Date 0214
- Bit #16 Conversion Date 0213
- Bit #32 Acquiring Institution Identification Code 018605
- Bit #33 Forwarding Institution Identification Code 018605
- Bit #37 Retrieval Reference Number 304510028931
- Bit #38 Approval Code 097044
- Bit #39 Response Code 00
- Bit #41 Card Acceptor Terminal Identification 30000791
- Bit #48 Additional Data, Private T4328jPmKUuKa+k/aCBEC65wEA5gAAAA=8701M660501011420701032129203847
- Bit #49 Transaction Currency Code 356
- Bit #50 Settlement Currency Code 840
- Bit #51 Cardholder Billing Currency Code 356
- Bit #56 ISO Reserved 013301295001D7D4TL3WMT6VZSAHA4VK48WKR
- Bit #63 Network Data MCB04WTDM



For Reference

- `com/fss/pg/iso/Type1ISOFormatter.java` - To Switch
- `com/fss/pg/iso/MASTERFormatter/TypeMasterISOFormatter.java` - To Mastercard
- `com/fss/pg/iso/VISAFormatter/TypeVisaISOFormatter.java` - To Visa



How is the interchange decided?

- Brand and BIN configuration

Issuer BIN

- First 6 digits of the Card Number
- Used to indicate the interchange (Card Networks such as Rupay, Mastercard, visa) and Issuer bank
- Given by the Interchange

BRAND

- Decides who is the interchange
- Bins are configured against BRAND

Standalone PG

- From – To BIN configured in Brand Configuration

Enable ON-US*

No



Authorization Entity

Switch

Description*

RUPAY

External Connection ID*

OTHERS BIC - Other Brand BIC



Payment Instruments*

☒ Credit Card

Details

☒ Debit Card

☐ 3D Secure

Details

☐ Prepaid Card

Details

BIN Ranges*

From BIN*

817200

To BIN*

819899

+

-

652154

652154



MAPS Integrated BRAND/BIN Config

- File with Bins uploaded in MAPS
- PG will receive the data and decide the BRAND based on the below
 - Acquiring Type – Onus/Offus
 - Card type – Debit/Credit/Prepaid
 - Internal/Domestic Cards
 - Consumer/Commercial Card

Example: IPY_CARDBIN_TABLE

ICD_LOW_RANGE – ICD_HIGH_RANGE

ICTH_SQID – Mastercard – 2/ Visa – 1 / Rupay - 5

CDTY_SQID – Debit - 2/Credit - 1/Prepaid - 3

PRO_CODE – Consumer- 1 /Commercial Card - 2

CNRY_SQID – International /Domestic Card

BRAND TYPE 1 – V, 2- M, 5 - N



3DS2.x

- Authentication protocol for card - based transactions
- OTP based authentication
- JSON Based

- **API**

- **PREQ/ PRES – Preparation Request / Response**
- Sent to Directory Server of the Card Network
- Used to get the list of bins enrolled in 3DS2
- Based on Start Range, End Range and ACS Info Indicator

- **AREQ/ARES - Authentication Request / Response**
- Sent to Directory Server of the Card Network
- Used to check whether the transaction is to be processed through Challenge Flow or Frictionless Flow
- Send RREQ and CRES Endpoint to DS
- Issuer ACS Endpoint to which CREQ must be sent



3DS2.x

- **RREQ/RRES - Results Request / Response**
 - Directory Server sends RREQ to PG
 - To retrieve CAVV and ECI
- **CREQ/CRES - Challenge Request / Response**
 - CREQ sent to Issuer ACS for OTP Authentication and Validation
 - Confirmation of Authentication in Challenge Flow



Frictionless vs Challenge flow

Frictionless Flow

- Does not require OTP Authentication and Validation
- Only AREQ-ARES is called
- Response is sent in ARES with Tran status and Authentication Value

Challenge Flow

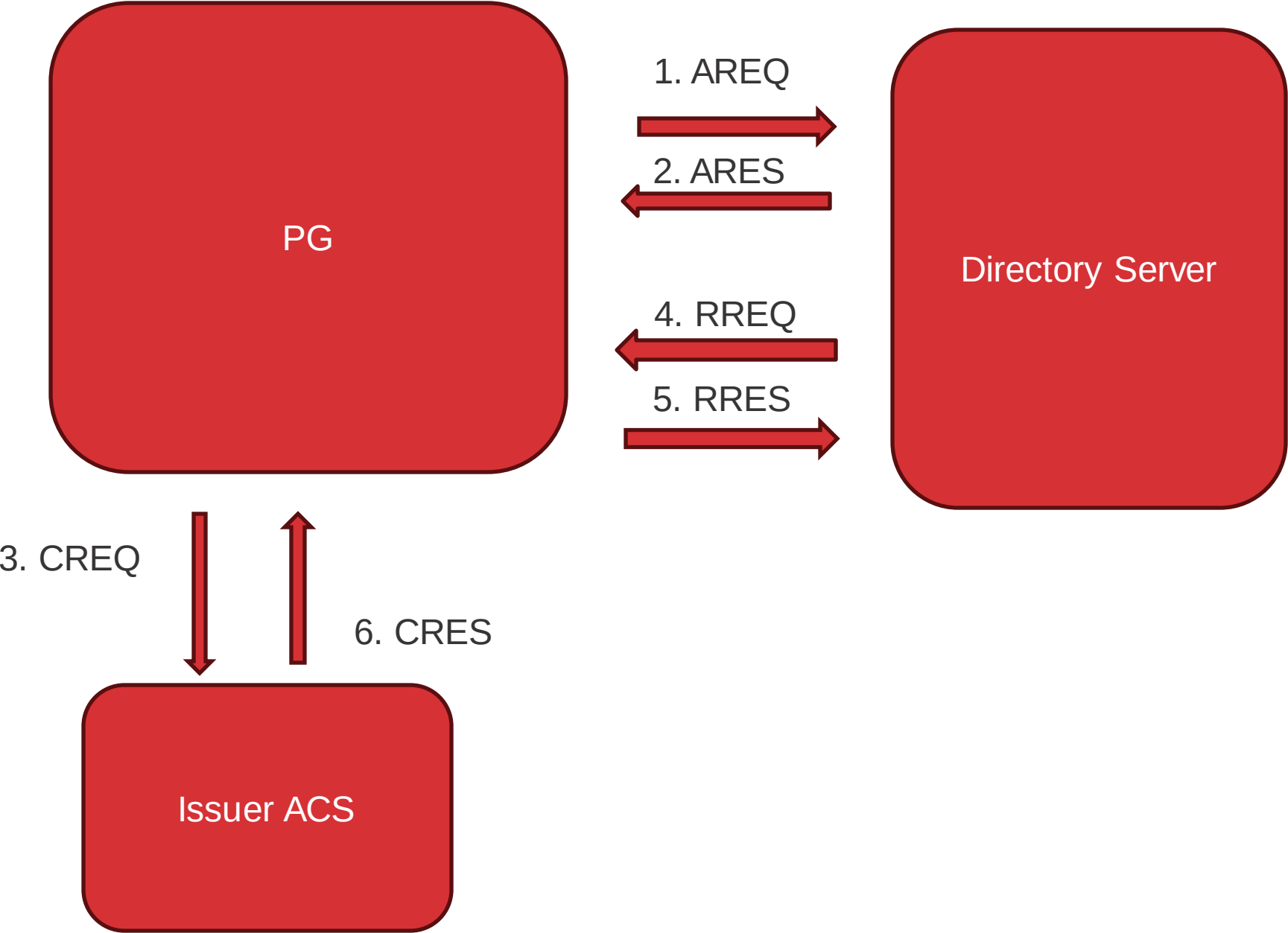
- Requires OTP authentication and Validation
- All 3DS2x APIs are called
- Response is sent in RREQ and CRES

How does the DS decide whether the transaction must go via Frictionless or Challenge Flow?

Risk check is done based on the Device and Browser data sent in AREQ, if the risk is low DS may route the transaction via frictionless flow

Acquirer PG has no role in deciding whether the transaction must go via Frictionless/challenge

3DS 2.x





3D Secure 2.x HTTP

1. Acquirer PG generates AREQ and sends it via S2S call to the directory server of the card network
2. DS validates the AREQ request received from Acquirer, and
 - Checks if merchant and acquirer is onboarded
 - Checks if card number is enrolled with the card network in 3DS2
 - Calculates risk, checks whether the transaction is to be initiated via Challenge or Frictionless flow
 - Endpoint where RREQ and CRES to be sent is sent in AREQ
3. ARES sent from DS to PG. From ARES
 - Endpoint where CREQ to be sent (Issuer ACS URL) is sent to PG
 - Confirmation on whether transaction is to be initiated via Challenge or Frictionless flow
4. CERQ to be sent Issuer ACS
5. OTP page will be opened in the browser for the customer to enter the OTP
6. OTP validation is done by the issuer ACS
7. RREQ is sent from DS to PG with **authenticationValue** , **tranStatus** and **ECI**
8. RRES is sent as an acknowledgement that we received RREQ
9. Finally, CRES is sent from Issuer to PG with **tranStatus**



3D Secure 2.x TCPIP

1. Acquirer PG generates AREQ and sends it via S2S call to the directory server of the card network
2. DS validates the AREQ request received from Acquirer, and
 - Checks if merchant and acquirer is onboarded
 - Checks if card number is enrolled with the card network in 3DS2
 - Calculates risk, checks whether the transaction is to be initiated via Challenge or Frictionless flow
 - Endpoint where RREQ and CRES to be sent is sent in AREQ
3. ARES sent from DS to PG. From ARES
 - Endpoint where CREQ to be sent (Issuer ACS URL) is sent to PG
 - Confirmation on whether transaction is to be initiated via Challenge or Frictionless flow
4. CREQ is sent to the Merchant
5. Merchant sends the CREQ to Issuer ACS
6. OTP page will be opened in the browser for the customer to enter the OTP
7. OTP validation is done by the issuer ACS
8. RREQ is sent from DS to PG with **authenticationValue** , **tranStatus** and **ECI**
9. RRES is sent as an acknowledgement that we received RREQ
10. CRES is sent to the Merchant from Issuer ACS
11. Finally, CRES is sent from Merchant to PG with **tranStatus**



AREQ – ARES

- Server 2 Server call
- SSL certificate validation

- {"threeDSCompInd":"N",
- "threeDSRequestorAuthenticationInd":"01",
- "threeDSRequestorID":"10020636*2000000000000303", **Configured in Merchant Level**
- "threeDSRequestorName":"Paytm Payments Ecom PT", **Merchant Name**
- "threeDSRequestorURL":"https://www.abcd.com", **Merchant Website Address**
- "threeDSServerRefNumber":"3DS_LOA_SER_FSSP_020200_00523", **Configured in Brand**
- "threeDSServerOperatorID":"10020636", **Configured in Brand**
- "threeDSServerTransID":"101c4339-5432-4b5d-877d-169cb411ee42", **Unique ID generated by the PG**
- "threeDSServerURL":"https://ipg.com/IPG/rReqHTTP.htm", **Endpoint to which RREQ must be sent**
- "acquirerBIN":"403274",
- "acquirerMerchantID":"2000000000000303", **Merchant ID**
- "browserAcceptHeader":"text/html,application/xhtml+xml,application/xml;q\u003d0.9,image/avif,image/webp,image/apng,*/*;q\u003d0.8,application/signed-exchange;v\u003db3;q\u003d0.9", **User Browser data**
- "browserIP":"172.34.**8.41", **User Browser data**
- "browserJavaEnabled":true, **User Browser data**
- "browserJavascriptEnabled":true, **User Browser data**
- "browserLanguage":"en-US", **User Browser data**
- "browserColorDepth":"24", **User Browser data**
- "browserScreenHeight":"864", **User Browser data**

- "browserScreenWidth":"1536", User Browser data
- "browserTZ":"300", User Browser data
- "browserUserAgent":"Mozilla/5.0 \u0026 #40;Linux; Android 13; I2127 Build/TP1A.220624.014; wv\u0026 #41; AppleWebKit/537.36 \u0026 #40;KHTML, like Gecko\u0026 #41; Version/4.0 Chrome/103.0.5060.129 Mobile Safari/537.36", User Browser data
- "cardExpiryDate":"2911", Card Expiry Date
- "acctNumber":"*****", Card Number
- email":"empty@abc.com, Customer Email Address
- "cardholderName":"*****", Card Holder Name
- "deviceChannel":"02",
- "merchantCountryCode":"356",
- "merchantName":"Paytm Payments Ecom PT",
- "messageCategory":"01",
- "messageType":"AReq",
- "messageVersion":"2.2.0",
- "notificationURL":"https://ipg.bobgateway.com/IPG/VPAS.htm?actionVPAS\u003dVbvCResProcessHTTP",
- "purchaseAmount":"199900",
- "purchaseCurrency":"356",
- "purchaseExponent":"2",
- "purchaseDate":"20230409090515",
- "transType":"01",
- "mcc":"5999"}



- {"threeDSServerTransID":"101c4339-5432-4b5d-877d-169cb411ee42","acsOperatorID":"3DS2.2.0Wib00497ACS","acsTransID":"b6ee8f7a-d687-11ed-a57e-41c47dd85e49","acsReferenceNumber":"3DS_LOA_ACS_WIBM_020200_00382","authenticationType":"02","acsChallengeMandated":"Y","dsReferenceNumber":"VISA.V 17 0003","dsTransID":"60b97798-533c-4693-b186-293968fa5c4b","messageType":"ARes","messageVersion":"2.2.0","transStatus":"C","acsURL":"https://secure-accs2ui-b4-indche-chnmdc.wibmo.com/v1/acs/services/browser/creq/L/8198/b6ee8f7a-d687-11ed-a57e-41c47dd85e49"}
- If transStatus = Y, Frictionless Flow successful
- TransStatus = C, Challenge Flow
- TransStatus = A , Attempted transaction
- TransStatus = N, Not Enrolled
- AcsURL – Issuer ACS URL where CREQ is to be sent
- DsTransID – Identifier at Interchange, **Sent in Authorization**
- acsTransID – Sent in CREQ

- CREQ is redirected via customer browser
- OTP page is opened after CREQ redirection
- CREQ is called only if Tran status = C in ARES / Challenge flow
- {"threeDSServerTransID":"101c4339-5432-4b5d-877d-169cb411ee42","acsTransID":"b6ee8f7a-d687-11ed-a57e-41c47dd85e49","challengeWindowSize":"03","messageType":"CReq","messageVersion":"2.2.0"}
- ThreeDSServerTransID – Unique ID sent in AREQ
- AcsTransID – From ARES



- Server to Server Call
- SSL Certificate Validation
- Sent from DS to PG
- PG is the server and DS acts as the client unlike other APIs
- ```
{ "threeDSServerTransID": "101c4339-5432-4b5d-877d-169cb411ee42", "acsTransID": "b6ee8f7a-d687-11ed-a57e-41c47dd85e49", "authenticationType": "02", "authenticationValue": "AAICBidhYwAAAwzcNWCZdQAAAAA=", "dsTransID": "60b97798-533c-4693-b186-293968fa5c4b", "eci": "05", "interactionCounter": "01", "messageCategory": "01", "messageType": "RReq", "messageVersion": "2.2.0", "transStatus": "Y" }
```
- AuthenticationValue / CAVV – to be sent in Authorization
- ECI – Sent in Authorization
- TranStatus – Status of authentication



- `{"threeDSServerTransID":"101c4339-5432-4b5d-877d-169cb411ee42","acsTransID":"b6ee8f7a-d687-11ed-a57e-41c47dd85e49","dsTransID":"60b97798-533c-4693-b186-293968fa5c4b","messageType":"RRes","messageVersion":"2.2.0","resultsStatus":"01"}`
- Acknowledge message to DS that we have received RREQ



- {"threeDSServerTransID":"101a6b25-687a-4e19-9372-b7b8ac88a2d3","acsTransID":"b3839f3b-d687-11ed-a18a-f576abbb76a0","messageType":"CRes","messageVersion":"2.2.0","challengeCompletionInd":"Y","transStatus":"Y"}
- ChallengeCompletionInd – Authentication done?
- TransStatus – Authentication successful ?
- TransStatus = Y, successful
- TransStatus = N, Failed